



Quantum Computing Technology

Exhaustive Research Report — 2025–2026 Updates

Prepared by Atlas for Finn McKie | 17 March 2026

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Executive Summary

Quantum computing crossed a decisive threshold in 2025–2026. What was once a physics experiment is now generating revenue, attracting sovereign-scale capital, and producing results that classical supercomputers genuinely cannot replicate. Here is what actually matters:

- Error correction works.** Quantinuum demonstrated 48 logical qubits from 98 physical qubits (2:1 ratio) on its Helios trapped-ion processor (March 2026). QuEra achieved 96 logical qubits on neutral atoms. Google's Willow chip proved that adding more qubits *reduces* errors — the first time this has been shown below threshold. The era of fault-tolerant quantum computing is no longer theoretical; it has a visible engineering path. [Sources: [Quantinuum](#), [QuEra](#)]
- Revenue is real.** IonQ became the first pure-play quantum company to surpass \$100M in annual revenue (FY2025). The global quantum computing market hit ~\$1.44–2.01 billion in 2025 and is projected at ~\$2 billion for 2026. Quantum computers sold: 37 units worth \$854M in 2024 alone. [Sources: [IonQ](#), [Yahoo Finance](#)]
- The geopolitical race is intensifying.** China's 15th Five-Year Plan (March 2026) explicitly targets quantum leadership with commitments to 1,000+ qubit control systems and a 12,000km quantum communication network. The EU launched its Quantum Europe

Strategy (July 2025) with €11B+ invested. The US leads in private capital but faces a coordination gap versus state-directed Chinese programmes.

4. **Microsoft's topological gambit is the wildcard.** Majorana 1 (February 2025) claims the first topological qubit — inherently more stable but unproven at scale. If it works, it could leapfrog everyone. If it doesn't, Microsoft has lost two decades. Nature reports physicists remain sceptical. [Source: [Nature](#)]
5. **Post-quantum cryptography migration is urgent.** NIST finalised three PQC standards (ML-KEM, ML-DSA, SLH-DSA) in August 2024. A US Executive Order in June 2025 mandated federal migration. "Harvest now, decrypt later" attacks are already underway. Every organisation should be migrating now.

My opinion: The most important development is not any single qubit count or funding round — it is that quantum error correction has been demonstrated to work *below threshold* across multiple architectures (superconducting, trapped-ion, neutral atom). This is the inflection point. Fault-tolerant quantum computing is now an engineering problem, not a physics problem. The companies that solve manufacturing and integration first — not necessarily those with the highest qubit counts — will dominate.

1. Hardware Milestones

1.1 Google — Willow Chip and Quantum Echoes

Google's Willow quantum chip (announced December 2024, results published through 2025) was the single most important hardware result of the period. Key achievements:

- **Below-threshold error correction:** For the first time, Google demonstrated that increasing the number of qubits in a surface code *reduced* the logical error rate rather than increasing it. This is the fundamental requirement for scalable quantum computing that the field has pursued for nearly 30 years. [Source: [Google Blog](#)]
- **Quantum Echoes algorithm (October 2025):** A new algorithm running on the 65-qubit Willow chip achieved a verified 13,000× speedup over the world's fastest supercomputer in a physics simulation task (OTOC measurements). Crucially, this was *verifiable* quantum advantage — not just a sampling benchmark but a computation whose results could be independently checked. Google called it "the first-ever algorithm to achieve verifiable quantum advantage on hardware." [Source: [Reuters](#)]
- **Roadmap:** Google is partnering with QuEra on neutral-atom systems and pursuing a dual-track roadmap combining superconducting and neutral-atom approaches.

1.2 IBM — Heron, Nighthawk, and the Utility Era

IBM has been the most consistent quantum hardware developer, operating 28 quantum computers with 100+ qubits (2,299 total available qubits as of March 2026) with 97% uptime.

- **Heron processor (2024–2025):** IBM's current workhorse, used in the landmark HSBC bond trading trial (September 2025) that delivered a 34% improvement in algorithmic bond price predictions. This was touted as the first empirical evidence of quantum computing value in financial services. [Source: [CBS News](#)]
- **Nighthawk processor (2025):** A higher-connectivity processor designed for more complex circuits. IBM describes it as "the brightest spot on the roadmap." It enables quantum + HPC hybrid workflows.
- **2026 roadmap target:** Demonstrate the first example of "scientific quantum advantage" and release a fault-tolerant prototype. IBM is targeting 5,000+ gate circuits on 156+ qubits with accurate execution. [Source: [IBM Quantum Roadmap](#)]
- **Revenue model:** IBM bundles hardware, cloud access, software (Qiskit), training, and consulting under multi-year contracts. They have run 3.6 trillion+ circuits to date.

1.3 Microsoft — Majorana 1 (Topological Qubits)

Microsoft unveiled Majorana 1 in February 2025 — its first quantum chip built on a topological architecture. This is radically different from every other approach:

- **Topological qubits** encode information in a non-local fashion using Majorana fermions, making them inherently more stable against noise. In theory, this could dramatically reduce the overhead needed for error correction.
- **The claim:** Microsoft says it has created the first topological qubits, published in *Nature* (February 2025). [Source: [Nature](#)]
- **The scepticism:** Multiple physicists remain unconvinced that Microsoft has truly demonstrated topological protection. The evidence is indirect, and the qubit counts are tiny compared to competitors.
- **Why it matters:** If topological qubits work as promised, they could be dramatically easier to scale than superconducting or trapped-ion approaches. Microsoft's roadmap targets a million-qubit machine. But the technology is at least 3–5 years behind competitors in practical demonstrations.
- **Strategic bet:** Microsoft is also partnering with Atom Computing (neutral atoms) as a hedge, suggesting even they are not fully confident in the topological-only path.

1.4 IonQ — Gate Fidelity World Record

IonQ set a world record in October 2025: **99.99% two-qubit gate fidelity** on trapped-ion hardware without ground-state cooling, using a "smooth gate" technique developed by Oxford Ionics (acquired by IonQ). [Source: [IonQ Investors](#)]

- **Why this matters:** Two-qubit gate errors are the primary bottleneck for quantum computation. Going from 99.9% to 99.99% is a 10× reduction in error rate, which exponentially reduces the overhead needed for error correction.
- **Roadmap:** 256-qubit prototype systems in 2026, millions of qubits by 2030. Logical qubits targeted for 2027–2028.
- **Financial milestone:** First quantum company to exceed \$100M annual revenue (FY2025). Total funding: \$3.6B+. Publicly traded on NYSE (IONQ). [Source: [Quantum Computing Report](#)]
- **Architecture:** Ytterbium ions, all-to-all connectivity (any qubit can interact with any other without shuttling). 64 physical qubits currently.

1.5 Quantinuum — Logical Qubit Leader

Quantinuum (Honeywell + Cambridge Quantum Computing merger, 2021) has emerged as arguably the most technically advanced quantum computing company:

- **Helios processor (launched 2025–2026):** 98 physical qubits (barium ions), QCCD architecture. Demonstrated **48 fully error-corrected logical qubits** with a remarkable **2:1 encoding ratio** using colour-code error correction. [Source: [The Quantum Insider](#)]
- **Beyond break-even (March 2026):** Quantinuum demonstrated that encoded circuits outperform their unencoded versions — meaning error correction is actively helping, not just adding overhead. Logical gate error rates of ~1 in 10,000 operations.
- **94 logical qubits demonstrated** in computations (March 2026), with 48 using concatenated codes.
- **DARPA QBI Stage B selected** alongside IonQ, confirming US government confidence in the approach.
- **IPO:** S-1 filed January 2026, targeting ~\$20B valuation. Total funding: \$900M+, valued at \$10B+.
- **HQ:** Cambridge, UK — making it the most valuable quantum company headquartered in Europe.

1.6 PsiQuantum — Photonic Approach

PsiQuantum announced **Omega** in February 2025 — a quantum photonic chipset published in Nature, purpose-built for million-qubit-scale quantum computers.

- Every photonic component demonstrated with beyond-state-of-the-art performance
- High-fidelity qubit operations and long-range chip-to-chip qubit interconnect (key scaling enabler)
- Manufactured using standard semiconductor fabrication (GlobalFoundries partnership)
- Strategy: skip NISQ entirely, go directly to fault-tolerant million-qubit machines

[Source: [Silicon UK](#)]

1.7 QuEra Computing — Neutral Atom Pioneer

QuEra (Harvard/MIT spinout) had a breakthrough year in 2025:

- **3,000-qubit array** sustained for over two hours of continuous operation, addressing the critical "atom loss" challenge
- **96 logical qubits** demonstrated with error rates that improved with system scale — the gold standard for fault-tolerant computing
- **\$230M+ in new capital** raised in 2025
- Partnered with Google on neutral-atom quantum computing
- Clear roadmap: 1,000 to million qubits via neutral-atom arrays

[Source: [QuEra via Entangled Future](#)]

1.8 Atom Computing

- Partnered with **Microsoft** to combine neutral-atom hardware with Microsoft's software stack
- Part of the neutral-atom consortium alongside QuEra and Pasqal driving the 1,000-to-million-qubit roadmap (2026–2036)
- Selected for DARPA programmes

1.9 Infleqtion

- Delivered the **UK's only operational 100-qubit quantum computing system** at the National Quantum Computing Centre (March 2026) using its SqaLe neutral-atom platform
- Listed on NYSE (INFQ)
- Key milestone for the UK's national quantum strategy

[Source: [Infleqtion](#)]

1.10 Architecture Comparison Summary

Architecture	Key Players	Physical Qubits (Best)	Logical Qubits (Best)	2-Qubit Fidelity	Scaling Path
Superconducting	Google, IBM	156 (IBM), 65 (Google Willow)	Below-threshold demonstrated (Google)	~99.5–99.8%	Modular linking, surface codes
Trapped Ion	IonQ, Quantinuum	98 (Quantinuum Helios)	48–94 (Quantinuum)	99.99% (IonQ record)	QCCD shuttling, modular
Photonic	PsiQuantum, Xanadu	N/A (different metric)	N/A (targeting million-scale)	High (Omega chipset)	Semiconductor fab, chip-to-chip
Topological	Microsoft	~8 (Majorana 1)	0 (not yet demonstrated)	Unknown	Inherent stability (if proven)
Neutral Atom	QuEra, Atom Computing, Infleqion, Pasqal	3,000 (QuEra array)	96 (QuEra)	~99.5%	Optical trapping, massive arrays

2. Software and Algorithm Breakthroughs

2.1 Google Quantum Echoes

The most significant algorithmic breakthrough of 2025. The Quantum Echoes algorithm: - Measures out-of-time-ordered correlators (OTOC) to reveal quantum behaviour - Achieved **13,000× speedup** over the world's fastest classical supercomputer - First *verifiable* quantum advantage on hardware (not just random circuit sampling) - Links directly to real-world applications including nuclear magnetic resonance (NMR) spectroscopy - Published October 2025

[Source: [Google Blog](#), [Reuters](#)]

2.2 Quantum Error Correction Codes

2025 was the year QEC became the universal priority:

- **Surface codes:** Google's below-threshold demonstration on Willow proved the theoretical framework works in practice

- **Colour codes:** Quantinuum's 2:1 encoding ratio is dramatically more efficient than surface codes (which require hundreds of physical qubits per logical qubit)
- **Iceberg error detection:** Used by Quantinuum for "beyond break-even" error-corrected circuits
- **Concatenated codes:** Quantinuum demonstrated 48 concatenated coded qubits (March 2026)

Riverlane (UK-based QEC specialist) identified three key trends for 2025: (1) investment/infrastructure maturation, (2) QEC as competitive differentiator, (3) hardware-software co-design for error correction. Their 2026 predictions: real-time decoding at scale, QEC-aware compilers becoming standard, and the first commercial QEC-protected workloads. [Source: [Riverlane Blog](#)]

2.3 Quantum Machine Learning

- **QuantumBoost (2025):** Google Quantum AI researchers developed a quantum boosting algorithm for machine learning, showing quantum speedups for ensemble learning. [Source: [arXiv](#)]
- **AdaBoost.Q:** Demonstrated on a programmable superconducting processor, improving prediction accuracy using the probabilistic nature of quantum measurement. Published in npj Quantum Information (May 2025). [Source: [Nature](#)]
- **Learning shallow quantum circuits:** New theoretical results on efficiently learning constant-depth quantum circuits, presented at COLT 2025. [Source: [PMLR](#)]
- **Hybrid quantum-classical:** Nvidia's Jensen Huang declared that connecting quantum computers directly to GPU supercomputers is "the future of quantum computing" — a major shift from his previous scepticism.

2.4 Programming Frameworks

- **Qiskit (IBM):** Remains the dominant open-source quantum SDK. Now integrated with HPC workflows via Nighthawk.
 - **Cirq (Google):** Used for Quantum Echoes and Willow experiments.
 - **TKET (Quantinuum/Cambridge Quantum):** Leading quantum compiler, now with QEC-aware optimisation.
 - **Amazon Braket, Azure Quantum:** Cloud platforms providing multi-vendor quantum access.
 - **CUDA Quantum (Nvidia):** Emerging framework for hybrid quantum-GPU computing.
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3. Commercial Applications and Industry

3.1 Finance — The First Killer App?

- **HSBC + IBM (September 2025):** 34% improvement in bond price predictions using IBM's Heron processor. Described as "ground-breaking world-first" and "the first piece of empirical evidence that quantum computing creates value in financial services." IBM's stock jumped 5% on the announcement. [Source: [Fortune](#)]
- **JPMorgan Chase + Quantinuum:** Achieved 100× improvement over existing industry benchmark using Quantinuum's H2-1 (Random Circuit Sampling). Ongoing partnership on portfolio optimisation using logical qubits.
- **Defence and aerospace** identified as the second key revenue sector for 2026.

3.2 Pharmaceuticals and Materials

- Drug discovery and molecular simulation remain the most cited long-term use case
- Still largely in the research/pilot phase — fault-tolerant machines needed for commercially relevant molecular simulations
- Key players: IBM (partnerships with Cleveland Clinic, Moderna), Google (DeepMind integration potential), Quantinuum (materials chemistry)

3.3 Market Size and Revenue

Metric	Value	Source
Global QC market 2025	\$1.44–2.01 billion	Precedence Research / InsightAce
Global QC market 2026 (projected)	~\$2 billion	Investing.com
Global QC market 2035 (projected)	\$19.44–40.45 billion	Precedence/InsightAce
CAGR 2026–2035	29.7–36.0%	Multiple sources
Quantum 2.0 market 2036 (projected)	\$50 billion	GlobeNewswire
QC companies revenue 2024	\$650–750 million	McKinsey
QC companies revenue 2025	\$1 billion+ (expected)	McKinsey
IonQ FY2025 revenue	\$100M+ (first to milestone)	IonQ
Quantum computers sold 2024	37 units, \$854M	Resonance
Average funding round 2025	\$50M+	ResearchAndMarkets

[Sources: [Yahoo Finance](#), [Investing.com](#), [BusinessWire](#)]

3.4 Investment Trends

- **First five months of 2025:** Global quantum investment reached ~75% of 2024's total with only ~25% of deal volume — meaning fewer rounds but much larger cheques. [Source: [QuantumBasel](#)]
 - **2024 total:** ~\$2 billion in investment, up 50% year-on-year from 2023
 - **Emerging startups:** Captured 37% of total 2024 funding; mature players drew 34% (McKinsey)
 - **Average order size:** Fell from \$48M (2021) to \$19M (2024) — market diversification
 - **Dedicated QC VC funds:** Quantonation, Quantum Coast Capital, increasingly joined by generalist VCs
 - **Key large raises:** QuEra (\$230M+), PsiQuantum (\$600M+ cumulative), IonQ (\$3.6B+ total)
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4. The Geopolitical Race

4.1 United States

- **Dominant in private capital:** Home to IonQ, Google Quantum AI, IBM Quantum, PsiQuantum, Atom Computing
- **DARPA Quantum Benchmarking Initiative (QBI):** Multi-stage programme targeting utility-scale quantum computing by 2033. IonQ and Quantinuum both selected for Stage B.
- **Executive Order (June 2025):** Mandated federal migration to post-quantum cryptography
- **National Quantum Initiative:** Renewed and expanded
- **Weakness:** Coordination gap — reliant on private sector versus China's state-directed model

4.2 China

China's quantum programme is the most comprehensive state-directed effort globally:

- **15th Five-Year Plan (March 2026):** Explicitly targets quantum leadership. AI mentioned 50+ times. Commits to:
 - Scalable quantum computers
 - 1,000+ qubit control systems with sub-microsecond feedback
 - Integrated space-earth quantum communication network
- 12,000km quantum network, 145 nodes, 17 provinces, 2 satellites already operational

- **Key institutions:** University of Science and Technology of China (USTC), led by Pan Jianwei
- **Key achievements:**
 - Jiuzhang 3.0 (photonic): quantum advantage in sampling problems
 - Zuchongzhi (superconducting): benchmarks directly against Google
 - World's largest quantum communication network
- **Company ecosystem:** Mostly state-backed spinouts from CAS labs and universities, not Silicon Valley-style startups. Key names include Origin Quantum, SpinQ, QuantumCTek.

[Sources: [The Quantum Insider](#), [Reuters](#), [Quantum Zeitgeist](#)]

4.3 European Union

- **Quantum Europe Strategy (July 2025):** Roadmap to make Europe a "quantum industrial powerhouse" by 2030
- **€11 billion+** in public funds invested in quantum R&D over five years (Quantum Flagship and related programmes)
- **Strengths:** World's largest pool of quantum researchers and startups; strong foundational science
- **Weakness:** Persistent gap translating research into commercial products ("Europe invents, America deploys, China scales")
- **Key companies:** Pasqal (France, neutral atom), IQM (Finland, superconducting), Alpine Quantum Technologies (Austria, trapped ion)
- **Quantinuum HQ:** Cambridge, UK — Europe's most valuable quantum company

[Source: [techUK](#), [PostQuantum.com](#)]

4.4 United Kingdom

- **National Quantum Computing Centre (NQCC):** Received UK's only operational 100-qubit system from Infleqtion (March 2026)
- **£2.5 billion National Quantum Strategy** (2023, 10-year plan)
- **Strengths:** Quantinuum HQ, Oxford Ionics (now IonQ), Riverlane (QEC), strong university ecosystem
- **Post-Brexit positioning:** Seeking to be the "quantum bridge" between US and EU

4.5 Investment by Country

Quantum computing and materials comprise ~70% of all quantum technology funding globally. The US and China lead strategic R&D, reflecting intense geopolitical involvement. [Source: [GlobeNewswire](#)]

5. Future Outlook

5.1 Fault-Tolerant Timeline

Based on current trajectories and expert statements:

Milestone	Estimated Timeline	Confidence
100+ logical qubits (demonstrated)	2026–2027	High
1,000 logical qubits	2028–2030	Medium
First commercial fault-tolerant workloads	2027–2029	Medium
Scientific quantum advantage (IBM target)	2026	Medium-High
Computationally useful quantum advantage (IBM)	2026	Medium
Million-qubit machines	2030–2035	Low-Medium
Cryptographically relevant quantum computer	2030–2040	Low (but "harvest now" threat is present)

5.2 Post-Quantum Cryptography

- **NIST standards finalised (August 2024):** ML-KEM (Kyber), ML-DSA (Dilithium), SLH-DSA (SPHINCS+)
- **US Executive Order (June 2025):** Federal migration mandated
- **"Harvest now, decrypt later" (HNDL):** Intelligence agencies in multiple countries actively collecting encrypted data for future quantum decryption. This is not theoretical. [Source: [CACM](#)]
- **Migration complexity:** Not a config change. Three distinct problems: key exchange, digital signatures, and protocol-level integration. Most organisations have not started. [Source: [Security Boulevard](#)]
- **Timeline pressure:** RSA-2048 could be broken by 2034 if quantum progress continues. Sensitive data encrypted today must be assumed compromised by then.

5.3 Quantum Internet and Networking

- **China leads:** 12,000km quantum communication network operational, 2 quantum satellites

- **Long-lived remote ion-ion entanglement (2026):** Published in Nature, demonstrating scalable quantum repeaters — a prerequisite for quantum internet. Achieved 1.2 million heralded Bell pairs at 11km. [Source: [Nature](#)]
- **Timeline:** Basic quantum key distribution networks operational now (China, some EU). Full quantum internet (distributed quantum computing, quantum sensor networks): 2035+

5.4 Impact on AI/ML

- **Hybrid quantum-classical:** The dominant paradigm. Nvidia's pivot to quantum-GPU integration validates this approach.
- **Near-term:** Quantum-enhanced optimisation, quantum-inspired algorithms running on classical hardware, quantum random number generation for AI training
- **Medium-term (2028–2032):** Quantum advantage in specific ML sub-tasks (kernel methods, Boltzmann machines, combinatorial optimisation)
- **Long-term:** Quantum simulation of molecular dynamics for drug discovery, materials science — areas where classical ML hits fundamental walls

5.5 What Matters Most — Atlas's Take

1. **Trapped ions are winning the quality race.** Quantinuum and IonQ have the best error rates and the most convincing paths to fault tolerance. Superconducting qubits (Google, IBM) win on quantity and ecosystem. Neutral atoms (QuEra) offer the best scaling potential. The winner will likely be hybrid.
 2. **Error correction is the game.** Raw qubit count is increasingly meaningless. What matters is logical qubits, encoding ratios, and error rates. Quantinuum's 2:1 ratio is extraordinary.
 3. **China is the only real competitor to the US.** The EU has excellent science but poor commercialisation. The UK punches above its weight via Quantinuum and the university ecosystem.
 4. **The PQC migration is the most urgent practical implication.** Organisations that haven't started are already behind. The threat is not when quantum computers break encryption — it's that data encrypted today is being harvested for future decryption.
 5. **2026–2028 is the critical window.** IBM says scientific quantum advantage in 2026. Multiple companies target commercial fault-tolerant workloads by 2028. If these timelines hold, quantum computing transitions from "promising technology" to "industrial tool" within Finn's current career horizon.
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